



**Università  
degli Studi  
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**Department of  
Computer Science and Artificial Intelligence**

**BIG DATA MANAGEMENT**

**Project Report**

**RESORT MANAGEMENT DATA WAREHOUSE AND BI  
(Data Warehouse Design and Business Intelligence Implementation)**

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## Abstract

This project presents the design and implementation of a Data Warehouse and Business Intelligence solution for a Resort Management System. The work was carried out using the Entity–Relationship (E/R) schema provided as the basis for the project and followed the requirements specified in the Big Data Management course.

The project began with the formulation of ten decisional queries representing the information needs of resort managers. These queries were used to identify candidate facts and guide the Data Warehouse design process. The original E/R schema was then refined through reification, resulting in a set of analytical entities suitable for dimensional modelling. From the identified facts, PAYMENT and ROOM\_RENTAL were selected for implementation because they provided the broadest coverage of the decisional queries.

Conceptual modelling was performed using the Dimensional Fact Model (DFM) methodology. Attribute trees were generated and subsequently refined through pruning and grafting. Dimensions, measures, and hierarchies were identified and used to construct the final fact schemas. The logical design phase produced Snowflake Schemas for the PAYMENT and ROOM\_RENTAL data marts.

The original operational database was implemented and populated using synthetic data generated with Python. The generated dataset was loaded into MySQL, and a SQL-based ETL process was developed to populate the Data Warehouse. The final warehouse was implemented in MySQL and consisted of shared dimensions and fact tables supporting the selected business processes.

To provide Business Intelligence capabilities, Metabase was connected to the Data Warehouse and used to develop dashboards and analytical reports. The resulting solution supports revenue analysis, accommodation performance analysis, customer demographic analysis, and seasonal reporting within a local offline environment.



## Chapter 1: Introduction

### 1.1 Project Overview

Data Warehouses play an important role in supporting decision-making by integrating data from operational systems and organizing it for analytical reporting. In resort management environments, managers often require information related to revenue, customer behaviour, accommodation performance, investments, discounts, and seasonal trends.

This project focuses on the design and implementation of a Data Warehouse and Business Intelligence solution for a Resort Management System based on the E/R schema provided in the project specification. The project follows the complete Data Warehouse development process, including decisional query analysis, fact identification, E/R schema refinement, conceptual design, logical design, database implementation, Data Warehouse implementation, and Business Intelligence reporting.

Among the candidate facts identified during the design phase, PAYMENT and ROOM\_RENTAL were selected for implementation. These facts were used to develop the data marts, analytical reports, and dashboards presented in this report.

### 1.2 Project Objectives

The objectives of the project are:

- Formulate decisional queries for the Resort Management System.
- Identify candidate facts from the decisional queries.
- Refine the original E/R schema through reification.
- Construct attribute trees for the selected facts.
- Apply pruning and grafting to obtain analytical structures.
- Identify dimensions, measures, and hierarchies.
- Design the final fact schemas.
- Develop Snowflake Schemas for the selected data marts.
- Implement and populate the Operational Database.
- Implement the Data Warehouse using SQL-based ETL processes.
- Develop dashboards and analytical reports using Metabase.

### 1.3 Tools and Technologies Used

Table 1.1 summarizes the tools and technologies used throughout the project.

**Table 1.1: Tools and Technologies Used**

| <b>Tool/Technology</b> | <b>Purpose</b>                                                                                              |
|------------------------|-------------------------------------------------------------------------------------------------------------|
| Draw.io                | Creation of reified schemas, attribute trees, pruned and grafted trees, fact schemas, and snowflake schemas |
| Python                 | Synthetic dataset generation                                                                                |
| MySQL                  | Operational Database and Data Warehouse implementation                                                      |
| SQL                    | Database creation, population, and ETL processing                                                           |
| MySQL Workbench        | Database development and administration                                                                     |
| Metabase               | Dashboard development and Business Intelligence reporting                                                   |

All components of the project were implemented in a local offline environment. Metabase was deployed using a self-hosted Java JAR installation and connected directly to the Data Warehouse.

## 1.4 Report Structure

The remainder of this report is organized as follows:

- **Chapter 2** presents the requirements analysis, original E/R schema, decisional queries, and candidate facts.
- **Chapter 3** presents the reification process and fact selection.
- **Chapter 4** presents the conceptual design, including attribute trees, pruning and grafting, dimensions, measures, and fact schemas.
- **Chapter 5** presents the logical design of the PAYMENT and ROOM\_RENTAL data marts.
- **Chapter 6** presents the implementation and population of the Operational Database.
- **Chapter 7** presents the Data Warehouse implementation and ETL process.
- **Chapter 8** presents the Business Intelligence dashboards and analytical reports.
- **Chapter 9** discusses the analytical results obtained from the implemented solution.
- **Chapter 10** concludes the project and outlines possible future enhancements.

## Chapter 2: Requirements Analysis

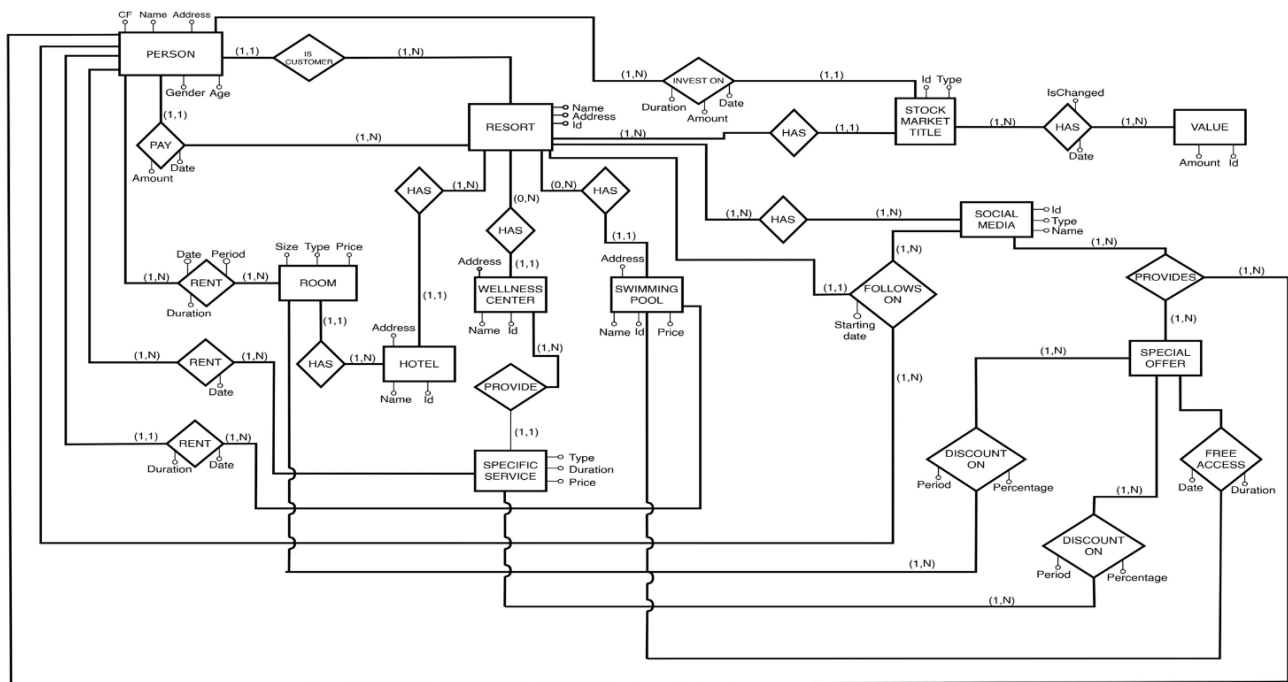
### 2.1 Overview

The first stage of the project focused on analysing the Resort Management System and identifying the information requirements that the Data Warehouse should support. This was achieved by examining the original E/R schema and formulating decisional queries that represent the information needs of resort managers.

The decisional queries served as the basis for identifying candidate facts and guided the subsequent conceptual and logical design activities.

### 2.2 Original E/R Schema

The project was based on the Resort Management System E/R schema provided in the assignment specification. The schema models the main operational activities of the resort, including customer payments, ROOM\_RENTALS, services, investments, discounts, social media interactions, and stock market activities.



**Figure 2.1: Original E/R Schema**

The original schema provided the foundation for identifying business processes that could support analytical reporting. These processes were later examined through decisional queries to determine the most suitable candidate facts for the Data Warehouse.

## 2.3 Decisional Queries

To identify the analytical requirements of the Resort Management System, ten decisional queries were formulated. These queries represent the type of information that managers may require when analysing resort operations and business performance.

**Table 2.1: Decisional Queries**

| No. | Query                                                                                                                                                         |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | What is the total and average revenue generated by customers from Italy during the month of August 2025?                                                      |
| 2   | During the Summer Business Season of 2025, which resort achieved the highest total ROOM_RENTAL revenue, and which room type contributed most to that revenue? |
| 3   | What is the total rental income generated by Suite room types within Hotel Alpha during the Summer Business Season of 2025?                                   |
| 4   | What is the total amount invested in Technology sector stock market titles by guests from France during 2025?                                                 |
| 5   | What is the count of new followers acquired on the Instagram platform during January 2025 from Italy?                                                         |
| 6   | What is the average discount percentage applied to Family ROOM_RENTALs during the Summer Business Season of 2025 in Hotel Europa?                             |
| 7   | What is the usage frequency and total revenue generated by the Massage service type for each month of 2025?                                                   |
| 8   | What is the average payment amount for Male customers in the 65+ Age Group from Italy during 2025?                                                            |
| 9   | What is the occupancy rate or rental utilization level for Double room types in Hotel Beta during the Winter Business Season of 2025?                         |
| 10  | How many value changes did AppleStock experience, and what was the average value variation per month during 2025?                                             |

The queries cover multiple business areas.

## 2.4 Facts Derived from Decisional Queries

Each decisional query was analysed to determine the business event responsible for generating the required information. The resulting candidate facts are summarized in Table 2.2.

**Table 2.2: Facts Derived from Decisional Queries**

| Query | Business Area                      | Main Fact   |
|-------|------------------------------------|-------------|
| Q1    | Revenue by customer geography/time | PAY         |
| Q2    | ROOM_RENTAL Performance Analysis   | RENT        |
| Q3    | Suite ROOM_RENTAL income           | RENT        |
| Q4    | Investment analysis                | INVEST ON   |
| Q5    | Social media followers             | FOLLOWS ON  |
| Q6    | Discount analysis                  | DISCOUNT ON |
| Q7    | Service revenue                    | RENT        |
| Q8    | Demographic payment analysis       | PAY         |
| Q9    | Room utilization                   | RENT        |
| Q10   | Stock value changes                | HAS         |

Table 2.2 shows the relationships support multiple decisional queries and represent the most significant business processes within the Resort Management System.

## 2.5 Facts Selection

The facts identified from the decisional queries were evaluated based on their analytical relevance and coverage of the business requirements.

The PAY relationship supports revenue analysis, geographical analysis, and demographic analysis through Queries Q1 and Q8.

The RENT relationship supports accommodation performance analysis, service revenue analysis, room utilization analysis, and rental income analysis through Queries Q2, Q3, Q7, and Q9.

Although other candidate facts were identified, PAY and RENT provided the broadest coverage of the decisional queries and were therefore selected for implementation in the Data Warehouse.

The remaining facts were retained during the conceptual design phase but were not implemented in the final Data Warehouse.

## 2.6 Summary

This chapter analysed the requirements of the Resort Management System using the original E/R schema and a set of decisional queries.

The next chapter refines the original E/R schema through reification and transforms the selected business processes into analytical entities suitable for dimensional modelling.

## Chapter 3: Fact Identification and E/R Refinement

### 3.1 Overview

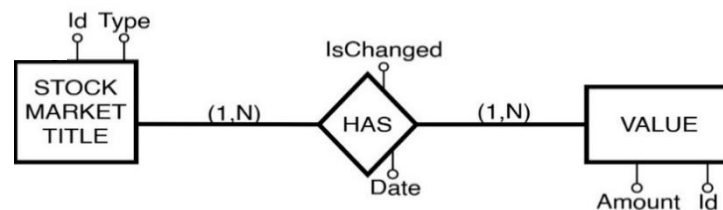
Following the requirements analysis presented in Chapter 2, the original E/R schema was refined to prepare it for dimensional modelling. This refinement focused on relationships representing business events that could support analytical reporting.

The refinement was performed through reification, resulting in a set of analytical entities that could be used during the conceptual design phase.

### 3.2 Reification Process

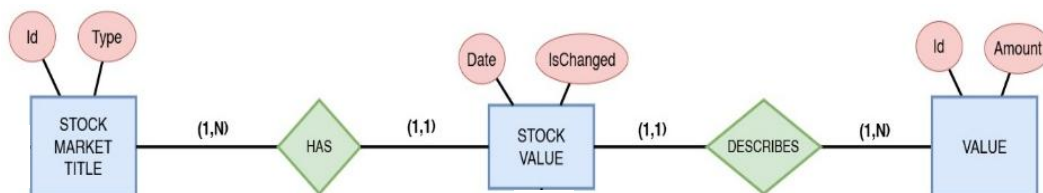
The candidate facts identified in Chapter 2 were originally represented as relationships in the operational E/R schema. To support dimensional modelling, these relationships were reified into analytical entities.

During reification, relationship attributes were transferred to the corresponding reified entity. New relationships were then introduced between the reified entity and the participating entities.



*Figure 3.1: Original HAS Relationship*

Figure 3.1 shows the HAS relationship before reification. The relationship contains the attributes Date and IsChanged and represents an association between STOCK MARKET TITLE and VALUE.



*Figure 3.2: Reified HAS Relationship*

After reification, the HAS relationship was transformed into the STOCK VALUE entity. The attributes Date and IsChanged became attributes of STOCK VALUE.

The original HAS relationship with cardinalities (1,N) and (1,N) was replaced by the STOCK VALUE entity. The new relationships, HAS and DESCRIBES, introduce a cardinality of (1,1) on the STOCK VALUE side, preserving the original business constraints represented in the E/R schema.

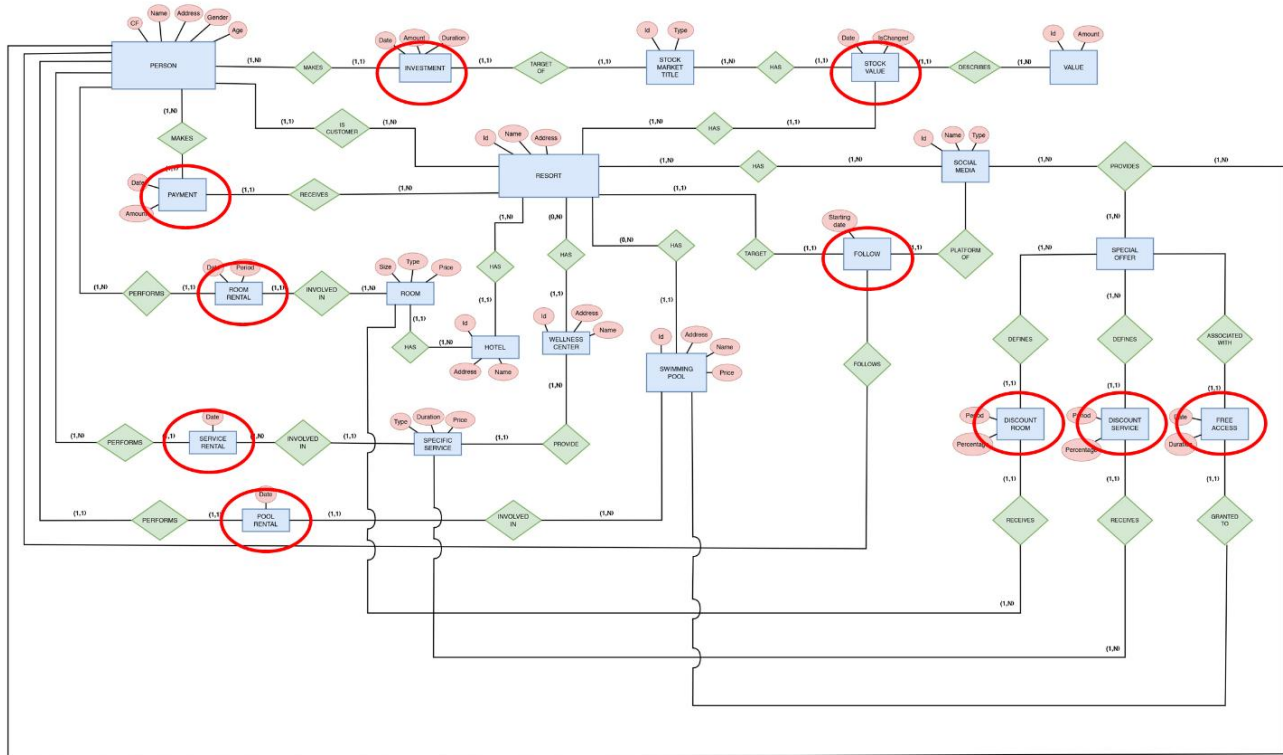
The same procedure was applied to all selected relationships in the schema.

### 3.3 Reified Entities

Table 3.1 summarizes the relationships selected for reification and their corresponding analytical entities.

**Table 3.1: Reification of Relationships into Analytical Entities**

| Relationship | Reified Entity   |
|--------------|------------------|
| PAY          | PAYMENT          |
| RENT         | ROOM_RENTAL      |
| RENT         | SERVICE RENTAL   |
| RENT         | POOL RENTAL      |
| INVEST ON    | INVESTMENT       |
| FOLLOWS ON   | FOLLOW           |
| DISCOUNT ON  | DISCOUNT ROOM    |
| DISCOUNT ON  | DISCOUNT SERVICE |
| HAS          | STOCK VALUE      |



**Figure 3.3: Reified E/R Schema**

The reified schema provides a clearer representation of the business events contained within the Resort Management System and forms the basis for conceptual modelling.

### 3.4 Fact Selection

The reified entities were evaluated against the decisional queries identified in Chapter 2.

PAYMENT supports:

- Q1 Revenue by customer geography and time
- Q8 Demographic payment analysis

ROOM\_RENTAL supports:

- Q2 ROOM\_RENTAL performance analysis
- Q3 Suite ROOM\_RENTAL income
- Q7 Service revenue analysis
- Q9 Room utilization analysis

Since PAYMENT and ROOM\_RENTAL support the largest number of decisional queries, they were selected for implementation in the Data Warehouse.



### **3.5 Summary**

This chapter refined the original E/R schema through reification and produced the analytical entities required for dimensional modelling. Based on the decisional queries, PAYMENT and ROOM\_RENTAL were selected for implementation. The next chapter develops the conceptual design through attribute trees, pruning and grafting, dimensions, measures, and fact schemas.

## Chapter 4: Conceptual Design

### 4.1 Overview

Conceptual modelling was performed for all reified entities identified in Chapter 3. Attribute trees were generated automatically and subsequently refined through pruning and grafting to identify dimensions, measures, and hierarchies suitable for dimensional modelling.

*For reporting purposes*, this chapter presents only the conceptual design of the PAYMENT and ROOM\_RENTAL facts, which were selected for implementation in the Data Warehouse. The same modelling procedure was applied to the remaining candidate facts during the design process.

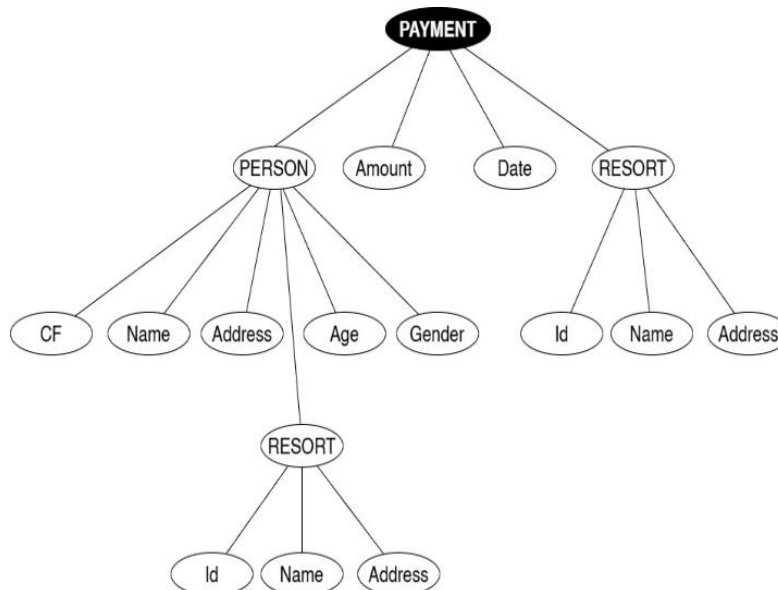
### 4.2 Pruning and Grafting

The attribute trees were refined through pruning and grafting. Pruning removed attributes that do not contribute to analytical reporting, while grafting reorganized and derived attributes required by the decisional queries. Both operations were performed together to ensure that the resulting structures were suitable for dimensional modelling and aligned with the analytical requirements.

### 4.3 PAYMENT Fact

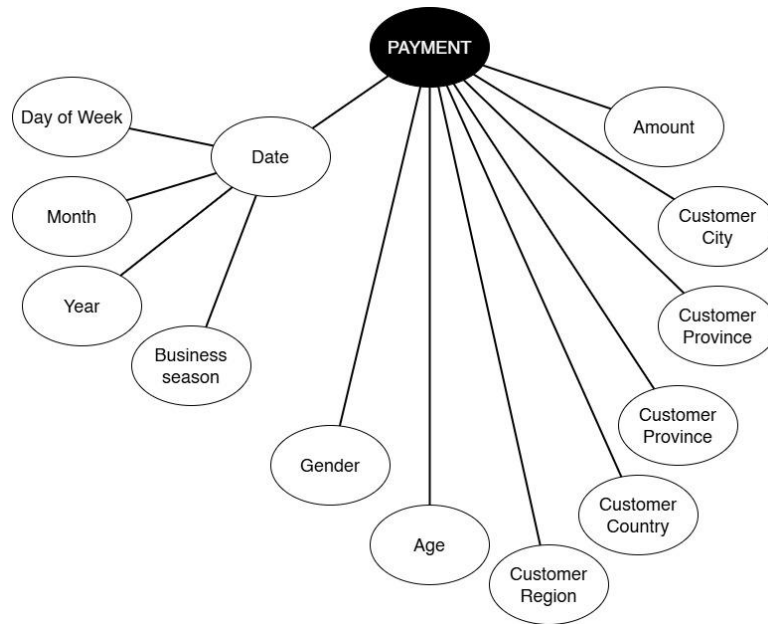
#### 4.3.1 PAYMENT Attribute Tree

Figure 4.1 shows the automatically generated attribute tree for the PAYMENT fact.



**Figure 4.1: PAYMENT Attribute Tree**

### 4.3.2 PAYMENT Pruned and Grafted Tree



**Figure 4.2: PAYMENT Pruned and Grafted Tree**

The PAYMENT attribute tree was refined to support revenue analysis across time, geography, and customer demographics.

**Table 4.1: PAYMENT Pruned and Grafted Attributes**

| Pruned Attributes | Grafted Attributes |
|-------------------|--------------------|
| CF                | Day of Week        |
| Customer Name     | Month              |
| Customer Address  | Year               |
| Resort ID         | Business Season    |
| Resort Name       | Customer City      |
| Resort Address    | Customer Province  |
|                   | Customer Region    |
|                   | Customer Country   |

### 4.3.3 PAYMENT Dimensions and Measures

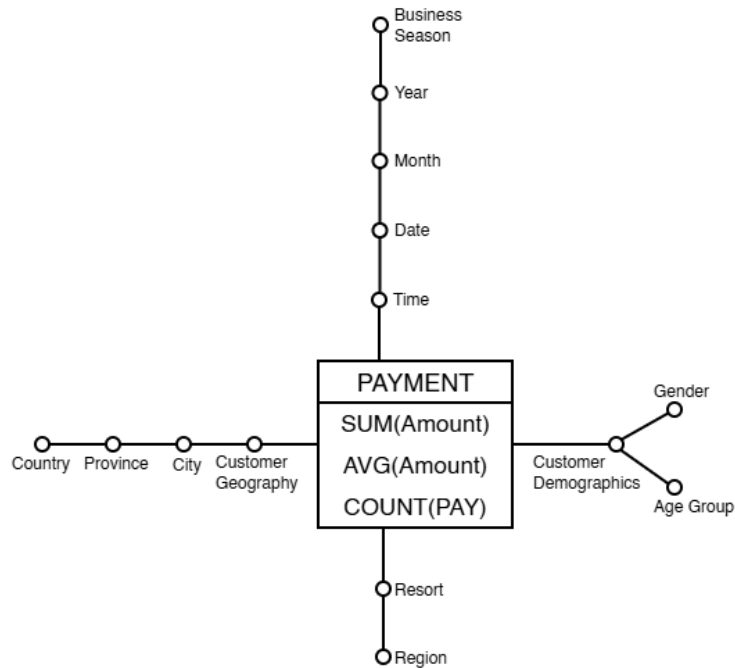
Dimensions and measures were selected from the pruned and grafted tree based on the requirements of Q1 and Q8 as shown in Table 4.2.

**Table 4.2: PAYMENT Dimensions and Measures**

| Measures                                 | Dimensions                                                                                                                                  |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| SUM(Amount)<br>AVG(Amount)<br>COUNT(PAY) | <b>Time</b>                                                                                                                                 |
|                                          | <ul style="list-style-type: none"> <li>• Date</li> <li>• Day of Week</li> <li>• Month</li> <li>• Year</li> <li>• Business Season</li> </ul> |
|                                          | <b>Customer Geography</b>                                                                                                                   |
|                                          | <ul style="list-style-type: none"> <li>• Customer City</li> <li>• Customer Province</li> <li>• Customer Country</li> </ul>                  |
|                                          | <b>Customer Demographics</b>                                                                                                                |
|                                          | <ul style="list-style-type: none"> <li>• Gender</li> <li>• Age Group</li> </ul>                                                             |
|                                          | <b>Resort</b>                                                                                                                               |
|                                          | <ul style="list-style-type: none"> <li>• Region</li> </ul>                                                                                  |

#### 4.3.4 PAYMENT Fact Schema

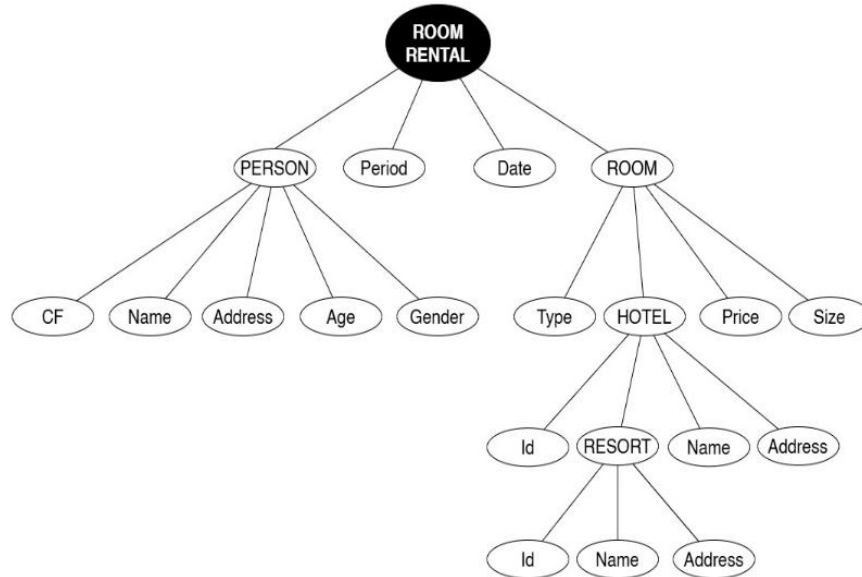
Figure 4.3 presents the final fact schema for PAYMENT.

**Figure 4.3: PAYMENT Fact Schema**

## 4.4 ROOM\_RENTAL Fact

### 4.4.1 ROOM\_RENTAL Attribute Tree

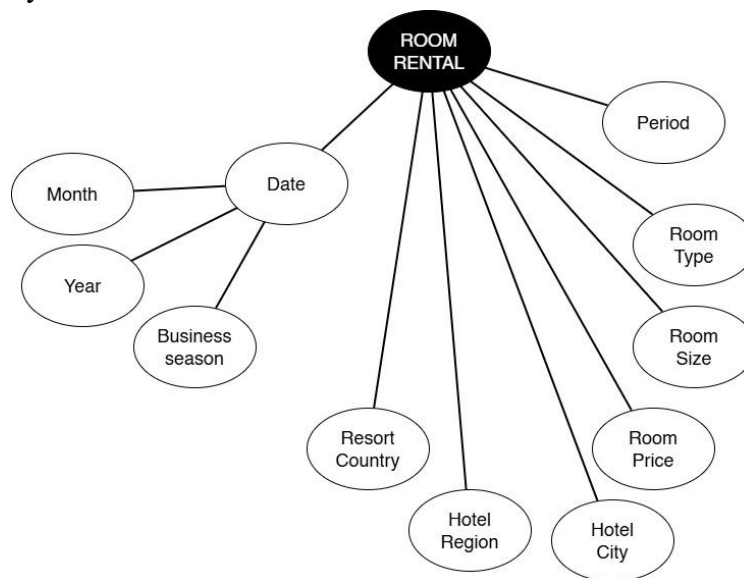
Figure 4.4 shows the automatically generated attribute tree for the ROOM\_RENTAL fact.



*Figure 4.4: ROOM\_RENTAL Attribute Tree*

### 4.4.2 ROOM\_RENTAL Pruned and Grafted Tree

The ROOM\_RENTAL attribute tree was refined to support accommodation performance and room utilization analysis.



*Figure 4.5: ROOM\_RENTAL Pruned and Grafted Tree*

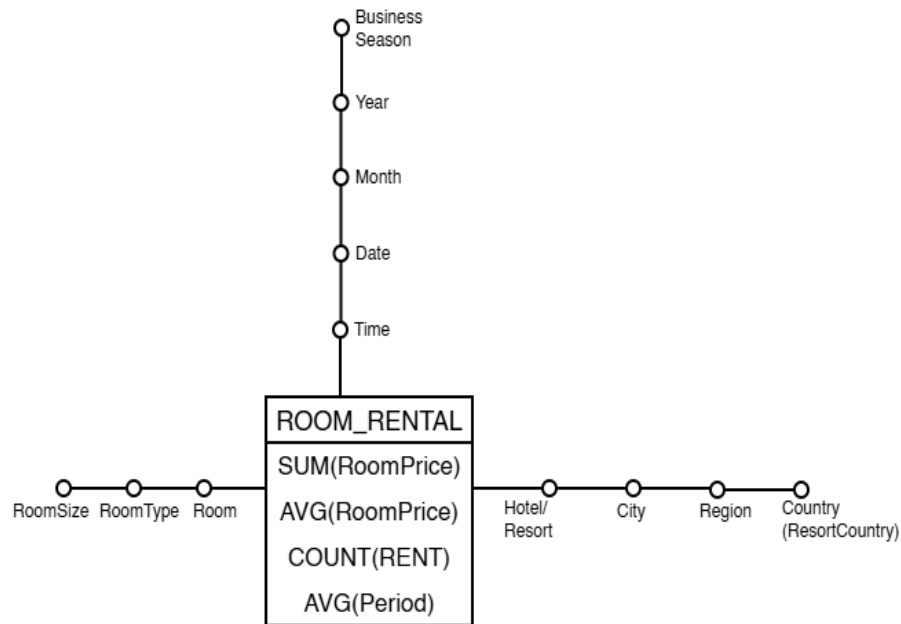
#### 4.4.3 ROOM\_RENTAL Dimensions and Measures

Dimensions and measures were selected from the pruned and grafted tree based on the requirements of Q2, Q3, Q7, and Q9.

**Table 4.2: ROOM\_RENTAL Dimensions and Measures**

| Measures                                                         | Dimensions                                                                                                                       |
|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| SUM(Room Price)<br>AVG(Room Price)<br>COUNT(RENT)<br>AVG(Period) | <b>Time</b> <ul style="list-style-type: none"> <li>• Date</li> <li>• Month</li> <li>• Year</li> <li>• Business Season</li> </ul> |
|                                                                  | <b>Room</b> <ul style="list-style-type: none"> <li>• Room Type</li> <li>• Room Size</li> </ul>                                   |
|                                                                  | <b>Hotel/Resort</b> <ul style="list-style-type: none"> <li>• City</li> <li>• Region</li> <li>• Country</li> </ul>                |

#### 4.4.4 ROOM\_RENTAL Fact Schema



**Figure 4.6: ROOM\_RENTAL Fact Schema**

## 4.5 Summary

This chapter presented the conceptual design of the PAYMENT and ROOM\_RENTAL facts. Attribute trees were generated and refined through pruning and grafting, resulting in analytical structures suitable for dimensional modelling. The resulting dimensions, measures, and hierarchies were then used to construct the final fact schemas. The next chapter presents the logical design of the corresponding data marts.

## Chapter 5: Logical Design

### 5.1 Overview

The logical design phase transformed the conceptual models developed in Chapter 4 into data marts suitable for Data Warehouse implementation. The identified dimensions, measures, and hierarchies were used to develop snowflake schemas for the PAYMENT and ROOM\_RENTAL facts.

### 5.2 Hierarchy Definition

The hierarchies were derived from the dimensions identified during conceptual design. These hierarchies support drill-down and roll-up operations during multidimensional analysis.

**Table 5.1: Selected Hierarchies**

| Hierarchy          | Levels                          |
|--------------------|---------------------------------|
| Time               | Date → Month → Year             |
| Customer Geography | City → Province → Country       |
| Resort Geography   | Resort → Region                 |
| Room               | Room Type → Room Size           |
| Hotel Geography    | Hotel → City → Region → Country |

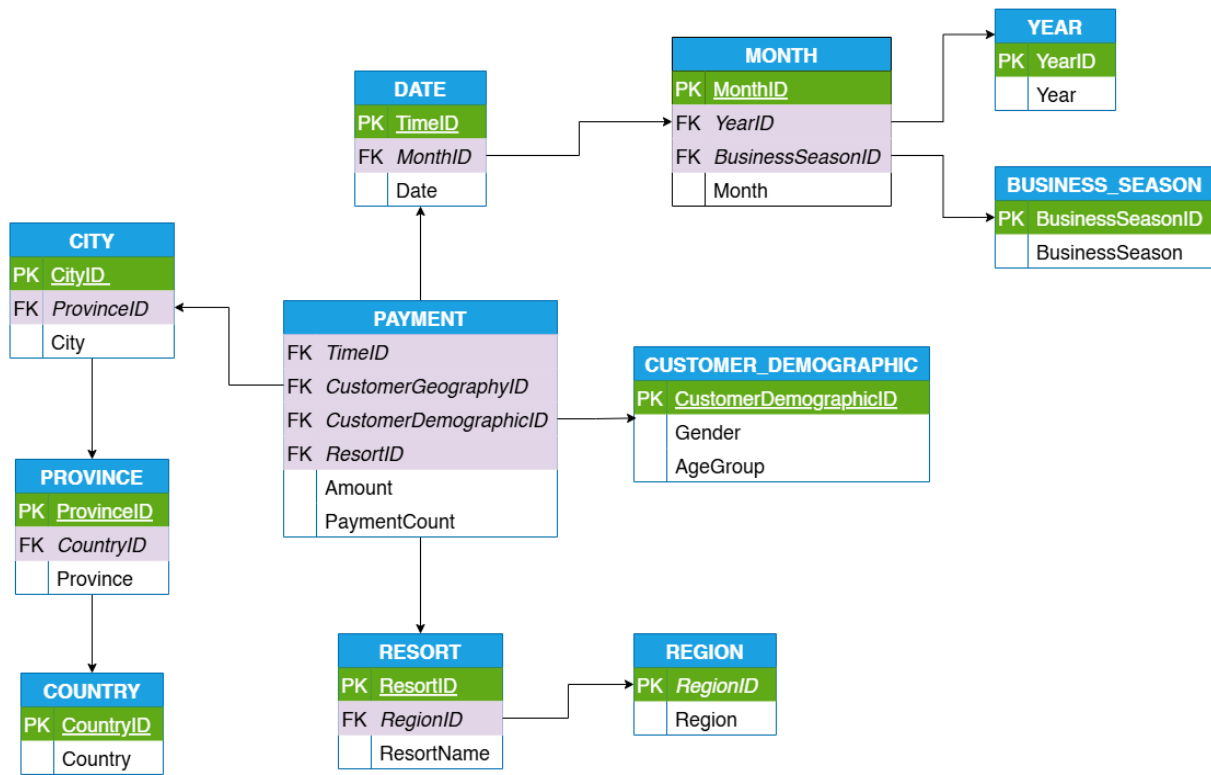
#### 5.2.1 Key Definition and Relationship Structure

Primary keys and foreign keys were identified during the logical design phase to support the implementation of the snowflake schemas.

- **Primary keys** were assigned to uniquely identify records within each dimension and fact table.
- **Foreign keys** were introduced to represent the relationships between hierarchy levels and to maintain referential integrity across the data marts.
- The hierarchy relationships identified during conceptual design were translated into linked dimension tables. For example, the Customer Geography hierarchy was implemented through the relationships: City → Province → Country
- Similarly, the Resort Geography, Hotel Geography, and Room hierarchies were implemented using foreign key relationships between successive hierarchy levels.



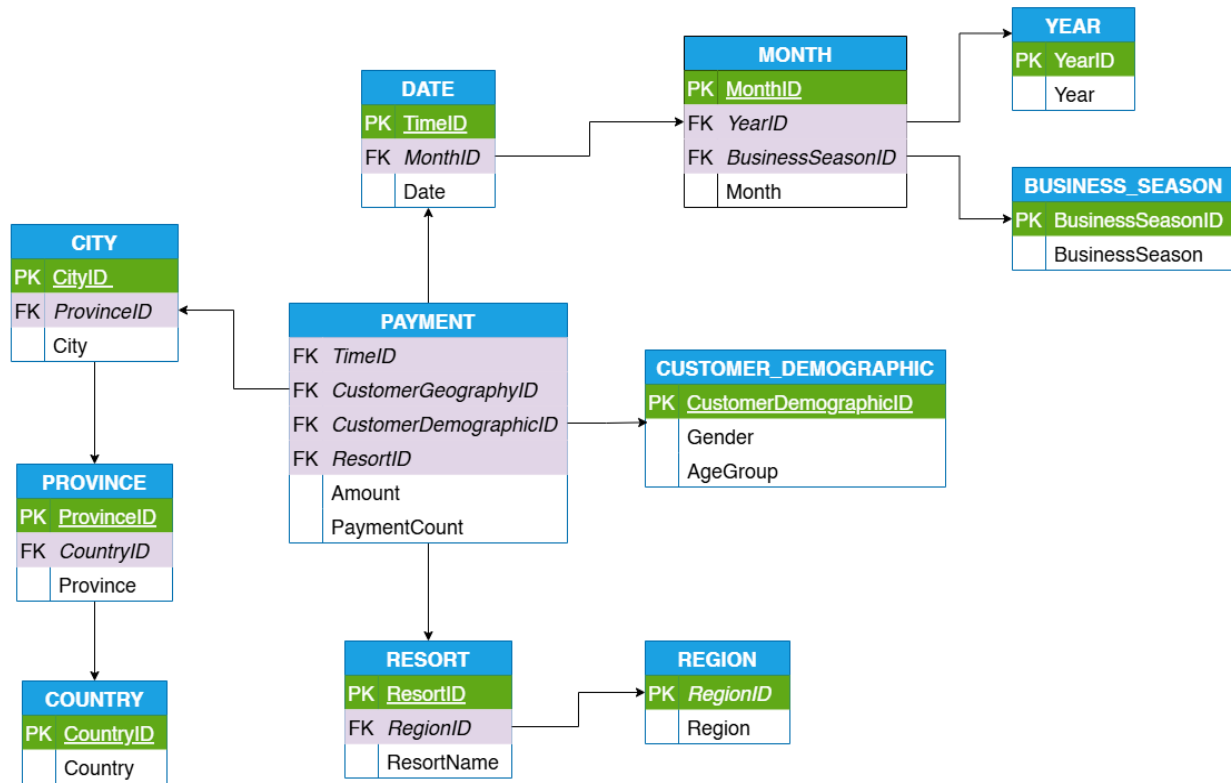
### 5.3 PAYMENT Data Mart



**Figure 5.1: PAYMENT Snowflake Schema**

The schema consists of the PAYMENT fact table and the associated dimensions for Time, Customer Geography, Customer Demographic, and Resort. Hierarchical relationships were normalized into separate dimension tables to support geographical and temporal analysis at multiple levels of aggregation.

## 5.4 ROOM\_RENTAL Data Mart



**Figure 5.2: ROOM\_RENTAL Snowflake Schema**

The ROOM\_RENTAL data mart was designed using a snowflake schema. The schema implements the dimensions, measures, and hierarchies identified during conceptual design and supports accommodation performance, rental revenue, and room utilization analysis.

## 5.5 Summary

This chapter presented the logical design of the PAYMENT and ROOM\_RENTAL data marts. The selected hierarchies were used to construct snowflake schemas that form the basis for the Data Warehouse implementation presented in the next chapter.

## Chapter 6: Operational Database Implementation

### 6.1 Overview

This chapter presents the implementation and population of the operational database based on the reified E/R schema developed in Chapter 3. The operational database was implemented in MySQL and populated using synthetic data generated with Python.

The implementation process consisted of three main stages:

- Dataset generation
- Database creation
- Database population

The resulting operational database served as the source system for the Data Warehouse implementation presented in Chapter 7.

### 6.2 Operational Database Design

The operational database was implemented from the reified E/R schema presented in Chapter 3. Figure 6.1 presents the implemented operational database schema which contains the primary entities; PERSON, RESORT, HOTEL, ROOM, PAYMENT, ROOM\_RENTAL, SERVICE\_RENTAL, POOL\_RENTAL, INVESTMENT, FOLLOW, DISCOUNT\_ROOM, DISCOUNT\_SERVICE, STOCK\_VALUE

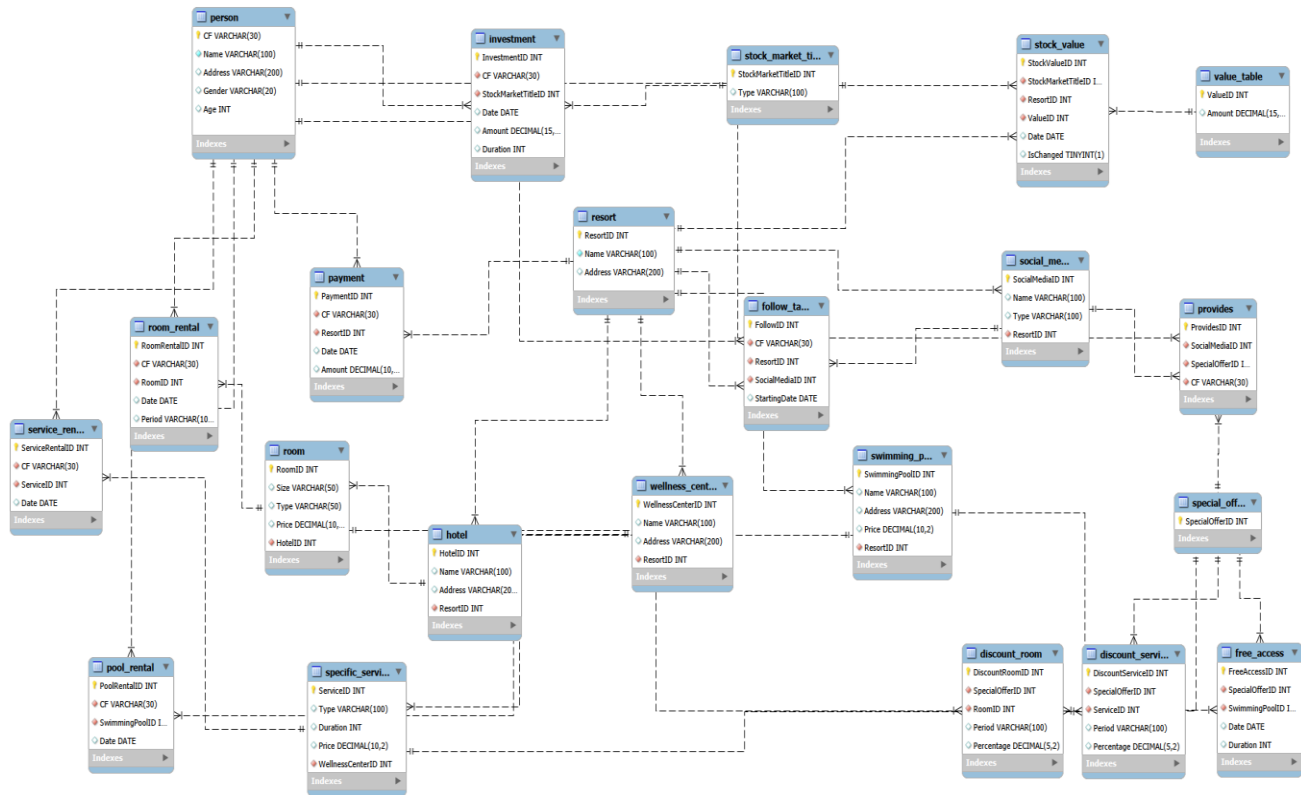
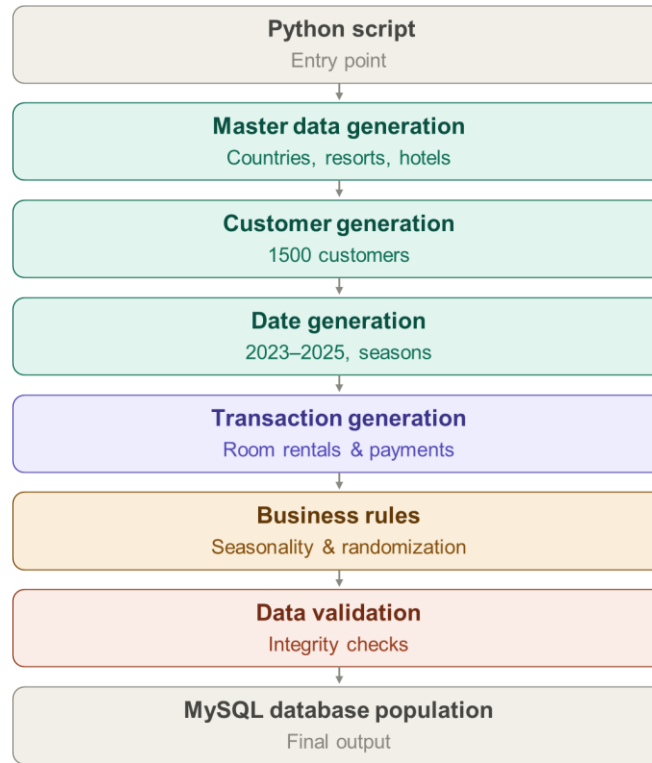


Figure 6.1: Operational Database Schema

### 6.3 Dataset Generation

Synthetic data was generated using Python to populate the operational database. The generated data was designed to simulate realistic resort operations while maintaining consistency with the implemented schema.



**Figure 6.2: Python Data Generation Flowchart**

#### 6.3.1 Python Libraries Used

**Table 6.1: Python Libraries Used**

| Library                              | Usage                                                   |
|--------------------------------------|---------------------------------------------------------|
| Random (python library)              | Generation of randomized business data                  |
| Datetime (python library)            | Creation and management of transaction dates            |
| collections.Counter (python library) | Validation and analysis of generated data distributions |

The Python libraries listed in Table 6.1 were used to generate random records, create dates, and validate generated data distributions.

```

1 import random
2 from datetime import date, timedelta
3 from collections import Counter
4

```

**Code Listing 6.3: Python Libraries Used**

### 6.3.2 Dataset Generation Parameters

The dataset generation process defined the number of records required for each entity in the operational database.

**Table 6.2: Generated Dataset Volumes**

| Entity             | Records Generated                                 |
|--------------------|---------------------------------------------------|
| PERSON             | 1,500                                             |
| RESORT             | 5                                                 |
| HOTEL              | 10                                                |
| ROOM               | 100                                               |
| WELLNESS_CENTER    | 8                                                 |
| SPECIFIC_SERVICE   | 35                                                |
| SWIMMING_POOL      | 15                                                |
| SOCIAL_MEDIA       | 15                                                |
| STOCK_MARKET_TITLE | 10                                                |
| VALUE_TABLE        | 150                                               |
| STOCK_VALUE        | 250                                               |
| SPECIAL_OFFER      | 60                                                |
| ROOM_RENTAL        | 900                                               |
| SERVICE_RENTAL     | 500                                               |
| POOL_RENTAL        | 300                                               |
| INVESTMENT         | 250                                               |
| FOLLOW             | 600                                               |
| DISCOUNT_ROOM      | 80                                                |
| DISCOUNT_SERVICE   | 70                                                |
| FREE_ACCESS        | 50                                                |
| PROVIDES           | 300                                               |
| PAYMENT            | Generated separately (1–10 payments per customer) |

```

1  import random
2  from datetime import date, timedelta
3  from collections import Counter
4
5  random.seed(42)
6
7  OUTPUT_FILE = "populate_resort_operational_db.sql"
8
9  # =====
10 # MAIN SETTINGS
11 # =====
12
13 PERSON_COUNT = 1500
14 RESORT_COUNT = 5
15 HOTEL_COUNT = 10
16 ROOM_COUNT = 100
17 WELLNESS_CENTER_COUNT = 8
18 SPECIFIC_SERVICE_COUNT = 35
19 SWIMMING_POOL_COUNT = 15
20 SOCIAL_MEDIA_COUNT = 15
21 STOCK_MARKET_TITLE_COUNT = 10
22 VALUE_TABLE_COUNT = 150
23 STOCK_VALUE_COUNT = 250
24 SPECIAL_OFFER_COUNT = 60
25
26 ROOM_RENTAL_COUNT = 900
27 SERVICE_RENTAL_COUNT = 500
28 POOL_RENTAL_COUNT = 300
29 INVESTMENT_COUNT = 250
30 FOLLOW_COUNT = 600
31 DISCOUNT_ROOM_COUNT = 80
32 DISCOUNT_SERVICE_COUNT = 70
33 FREE_ACCESS_COUNT = 50
34 PROVIDES_COUNT = 300
35
36 # PAYMENT is generated separately:
37 # each person makes 1 to 10 payments.
38

```

**Code Listing 6.2: Dataset Generation Parameters**

### 6.3.3 Seasonal Data Distribution

To simulate realistic resort activity, the generated dataset incorporated seasonal behaviour. Records were generated between 2023 and 2025, with a higher probability assigned to the summer months.

```

196 def weighted_tourism_date():
197     """
198     Generates dates between 2023 and 2025.
199     Summer months have the highest probability.
200     """
201     year = random.choice([2023, 2024, 2025])
202
203     month = random.choices(
204         population=[1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12],
205         weights=[3, 3, 4, 6, 8, 14, 18, 18, 10, 7, 5, 4],
206         k=1
207     )[0]
208

```

**Code Listing 6.3: Summer Season Bias Logic**

The seasonal weighting increased the frequency of transactions occurring during June, July, and August.

### 6.3.4 Date Generation Logic

Date values used throughout the generated dataset were randomly selected within the defined operational period.

```
201 | year = random.choice([2023, 2024, 2025])
```

*Code Listing 6.4: Date Generation Logic*

The generated dates support temporal analysis across multiple years and seasonal periods.

### 6.4 Database Creation

The operational database was created in MySQL using SQL Data Definition Language (DDL) statements. Tables, primary keys, foreign keys, and integrity constraints were defined according to the implemented schema.

```
7 CREATE DATABASE IF NOT EXISTS resort_operational_db;
8
9 USE resort_operational_db;
10
11 -- =====
12 -- 1. PERSON
13 -- =====
14
15 CREATE TABLE PERSON (
16 CF VARCHAR(30) PRIMARY KEY,
```

*Code Listing 6.5: Database Creation Script Excerpt*

The complete database creation script is available in the project GitHub repository (Appendix A).

### 6.5 Database Population

After creating the database structure, the generated dataset was loaded into the corresponding tables. The population process maintained referential integrity between related entities and relationships.

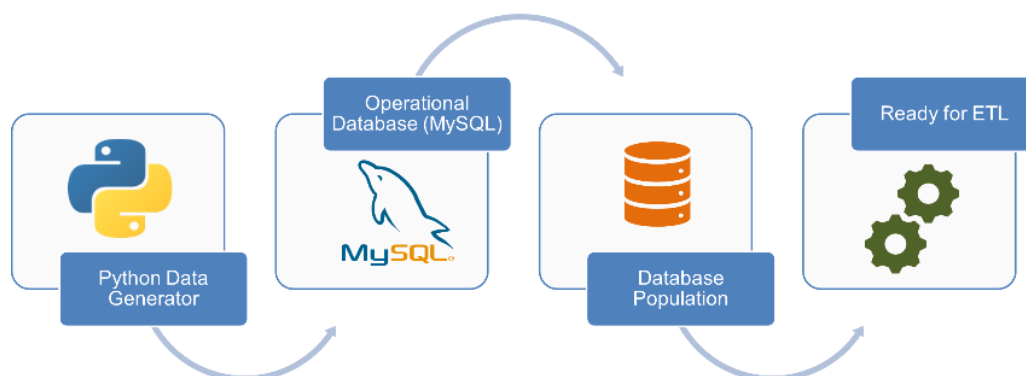
```
2184 INSERT INTO PAYMENT (PaymentID, CF, ResortID, `Date`, Amount) VALUES (1, 'CF00001', 1, '2024-08-31', 1430.11);
2185 INSERT INTO PAYMENT (PaymentID, CF, ResortID, `Date`, Amount) VALUES (2, 'CF00001', 4, '2024-09-20', 497.92);
2186 INSERT INTO PAYMENT (PaymentID, CF, ResortID, `Date`, Amount) VALUES (3, 'CF00001', 3, '2025-10-11', 838.83);
2187 INSERT INTO PAYMENT (PaymentID, CF, ResortID, `Date`, Amount) VALUES (4, 'CF00001', 3, '2025-07-20', 338.72);
2188 INSERT INTO PAYMENT (PaymentID, CF, ResortID, `Date`, Amount) VALUES (5, 'CF00001', 3, '2024-09-01', 522.09);
2189 INSERT INTO PAYMENT (PaymentID, CF, ResortID, `Date`, Amount) VALUES (6, 'CF00001', 4, '2025-09-30', 242.20);
2190 INSERT INTO PAYMENT (PaymentID, CF, ResortID, `Date`, Amount) VALUES (7, 'CF00001', 5, '2023-10-09', 956.93);
```

*Code Listing 6.6: Database Population Script Excerpt*

The complete database population script is available in the project GitHub repository (Appendix A).

Successful population of the operational database provided the source data required for the extraction, transformation, and loading processes described in Chapter 7.

The workflow adopted in this project is shown below:



**Figure 6.3: SQL Database Population Screenshot**

After execution of the population script, the Operational Database contained the complete dataset required for the ETL process and subsequent Data Warehouse implementation.

## 6.6 Summary

This chapter presented the implementation and population of the operational database. The operational schema was implemented in MySQL and populated using synthetic data generated with Python. The resulting database provides the operational data source used for the Data Warehouse implementation presented in the next chapter.

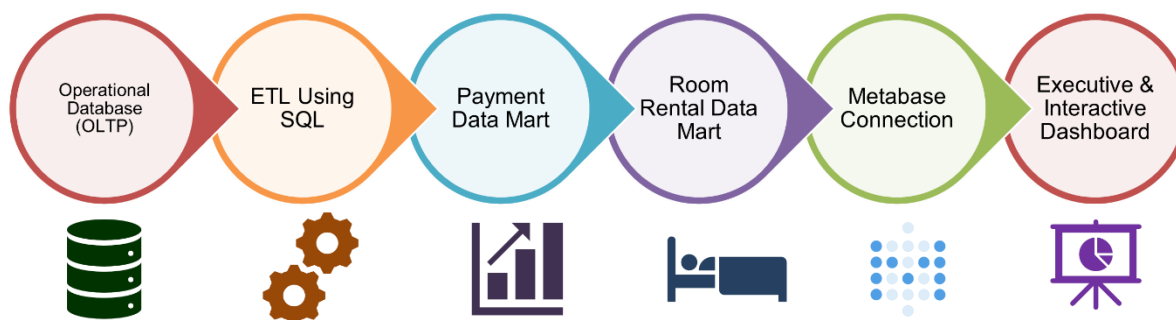


## Chapter 7: Data Warehouse Implementation

### 7.1 Overview

This chapter presents the implementation of the Data Warehouse and the ETL process used to transform operational data into analytical structures. The warehouse was implemented in MySQL within a dedicated database named `resort_dw` and supports the PAYMENT and ROOM\_RENTAL business processes selected during the design phase.

### 7.2 Data Warehouse Workflow



*Figure 7.1: Data Warehouse Implementation Workflow*

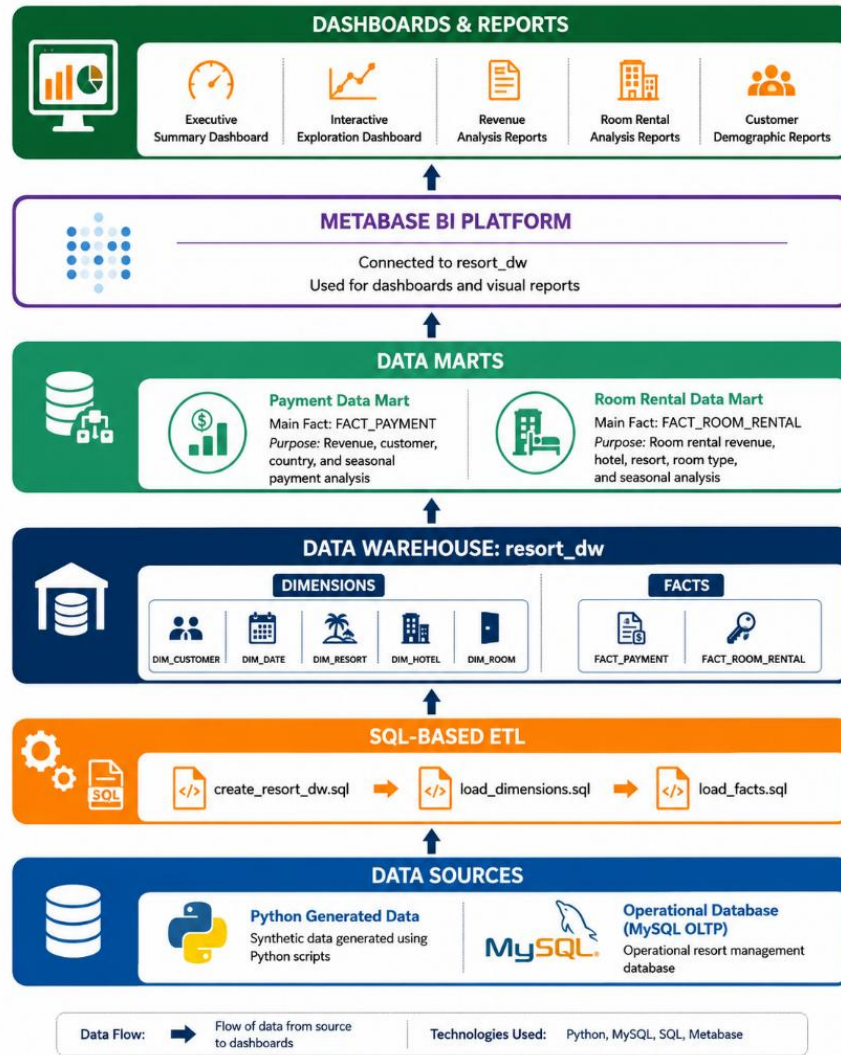
Figure 7.1 illustrates the workflow adopted in the project. Data generated and stored in the Operational Database was extracted, transformed, and loaded into the Data Warehouse using SQL scripts. The resulting warehouse served as the source for the Payment and ROOM\_RENTAL Data Marts and the Business Intelligence environment implemented in Metabase.

The entire solution was deployed in a local offline environment, including the Operational Database, Data Warehouse, ETL scripts, and self-hosted Metabase installation.

### 7.3 Data Warehouse Architecture

The implemented architecture consists of the Operational Database, SQL-based ETL layer, Data Warehouse (`resort_dw`), analytical Data Marts, and the Metabase reporting layer.

The architecture follows established Data Warehouse principles separating operational and analytical processing (Golfarelli & Rizzi, 2009).



**Figure 7.2: Data Warehouse Architecture**

The warehouse contains five dimension tables:

- DIM\_DATE
- DIM\_CUSTOMER
- DIM\_RESORT
- DIM\_HOTEL
- DIM\_ROOM

and two fact tables:

- FACT\_PAYMENT
- FACT\_ROOM\_RENTAL

#### 7.4 Data Warehouse Creation

The Data Warehouse was created using the create\_resort\_dw.sql script.

```

1  DROP DATABASE IF EXISTS resort_dw;
2  CREATE DATABASE resort_dw;
3  USE resort_dw;
4
5  CREATE TABLE DIM_DATE (
6      DateID INT AUTO_INCREMENT PRIMARY KEY,
7      FullDate DATE NOT NULL UNIQUE,

```

**Code Listing 7.1: Data Warehouse Creation Script Excerpt**

The script created the resort\_dw database together with the required dimension and fact tables defined during the logical design phase.

## 7.5 ETL Implementation

ETL processes are responsible for extracting data from operational systems, transforming it into analytical structures, and loading it into a Data Warehouse (Inmon, 2005).

The ETL process was implemented using SQL scripts executed in the following sequence:

1. create\_resort\_dw.sql
2. load\_dimensions.sql
3. load\_facts.sql

This sequence ensured that all dimension records were available before loading fact records.

### 7.5.1 Dimension Loading

The load\_dimensions.sql script populated the dimension tables from the Operational Database.

```

1  USE resort_dw;
2
3  INSERT INTO DIM_DATE
4  (FullDate, DayOfWeek, MonthNumber, MonthName, QuarterNumber, YearNumber, BusinessSeason)
5  SELECT DISTINCT
6      d.FullDate,
7      DAYNAME(d.FullDate),
8      MONTH(d.FullDate),

```

**Code Listing 7.2: Dimension Loading Script Excerpt**

**Table 7.1: Key transformations performed during dimension loading**

| DIM_DATE        | DIM_CUSTOMER | DIM_ROOM   |
|-----------------|--------------|------------|
| Month Name      | Age Group    | Room Type  |
| Business Season | City         | Room Size  |
| Day of Week     | Country      | Room Price |
| Quarter Number  |              |            |
| Month Number    |              |            |
| Year Number     |              |            |

### 7.5.2 Fact Loading

The load\_facts.sql script populated FACT\_PAYMENT and FACT\_ROOM\_RENTAL.

```

1  USE resort_dw;
2
3  INSERT INTO FACT_PAYMENT
4  (SourcePaymentID, DateID, CustomerDimID, ResortDimID, Amount, PaymentCount)
5  SELECT
6      p.PaymentID,
7      dd.DateID,
8      dc.CustomerDimID,
9      dr.ResortDimID,
10     p.Amount,

```

*Code Listing 7.3: Fact Loading Script Excerpt*

- FACT\_PAYMENT stores payment transactions together with the Amount measure and PaymentCount indicator.
- FACT\_ROOM\_RENTAL stores rental transactions together with RentalDays and RentalRevenue measures.

RentalRevenue was calculated as:

$$\text{Rental Revenue} = \text{Room Price} \times \text{Rental Days}$$

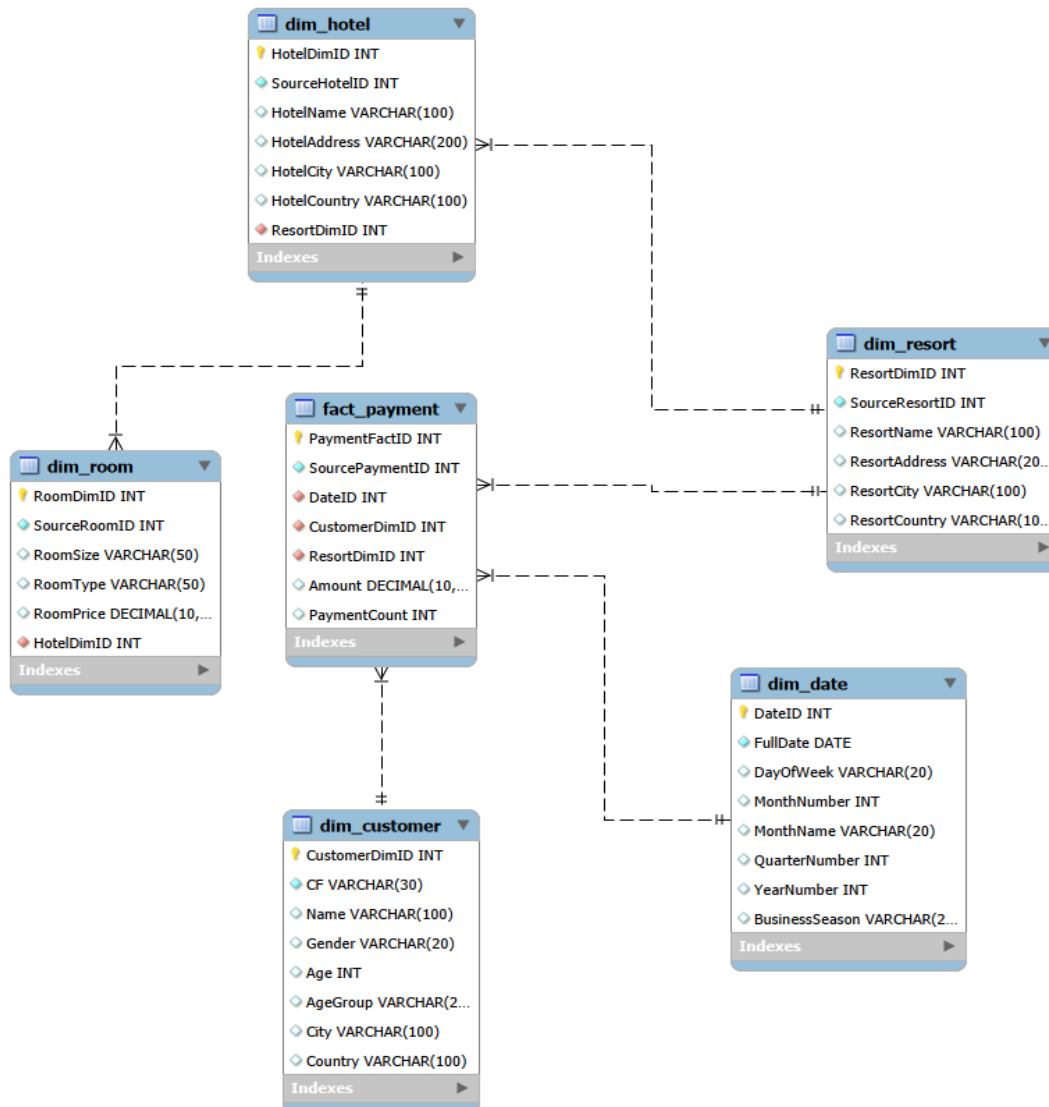
where RentalDays was derived from the rental period stored in the Operational Database.

## 7.6 Implemented Data Marts

### Payment Data Mart

The Payment Data Mart is centred on FACT\_PAYMENT and supports:

- Revenue Analysis
- Customer Demographic Analysis
- Geographic Analysis
- Seasonal Payment Analysis

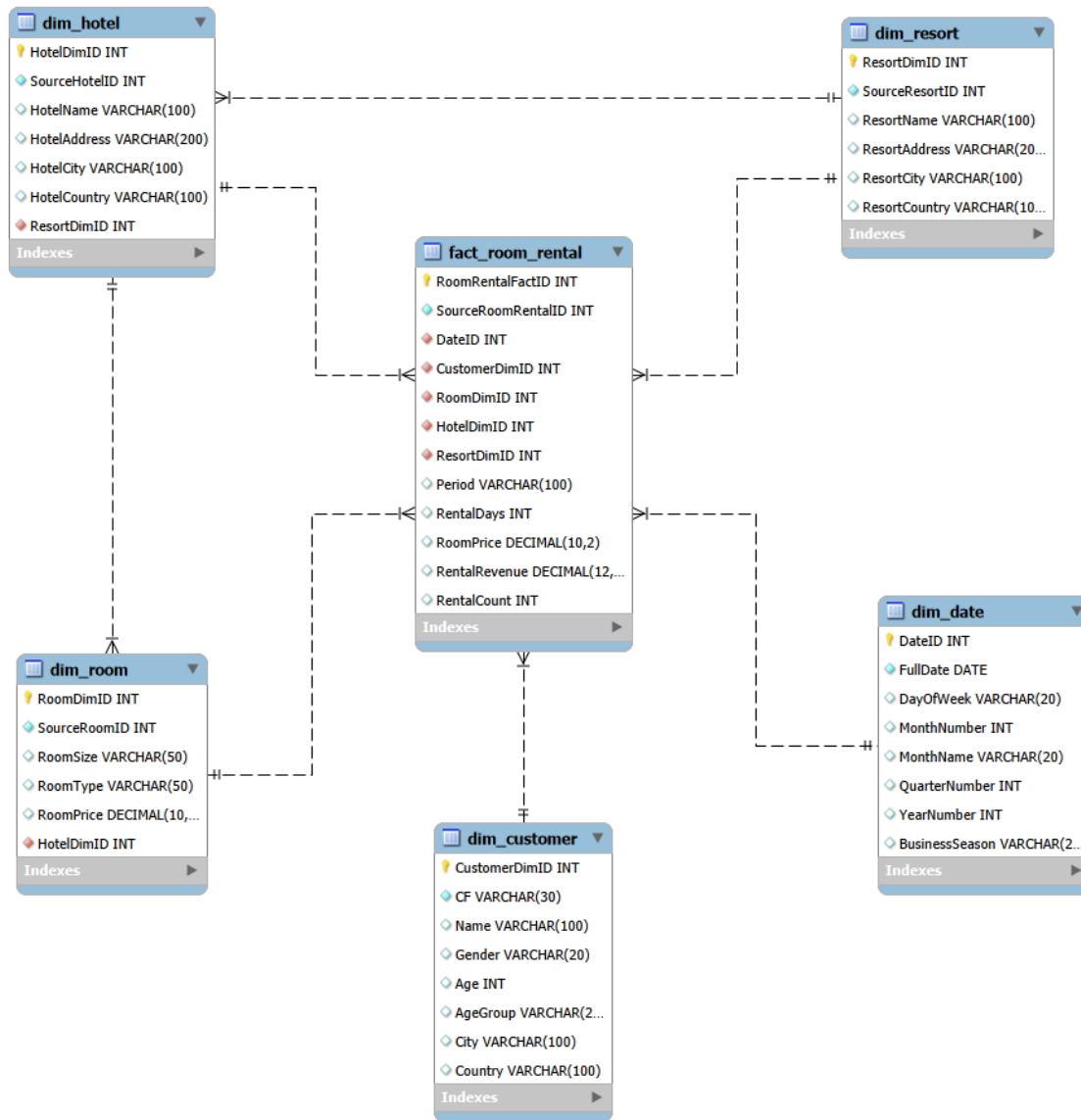


*Figure 7.3: Payment Data Mart*

### ROOM\_RENTAL Data Mart

The ROOM\_RENTAL Data Mart is centred on FACT\_ROOM\_RENTAL and supports:

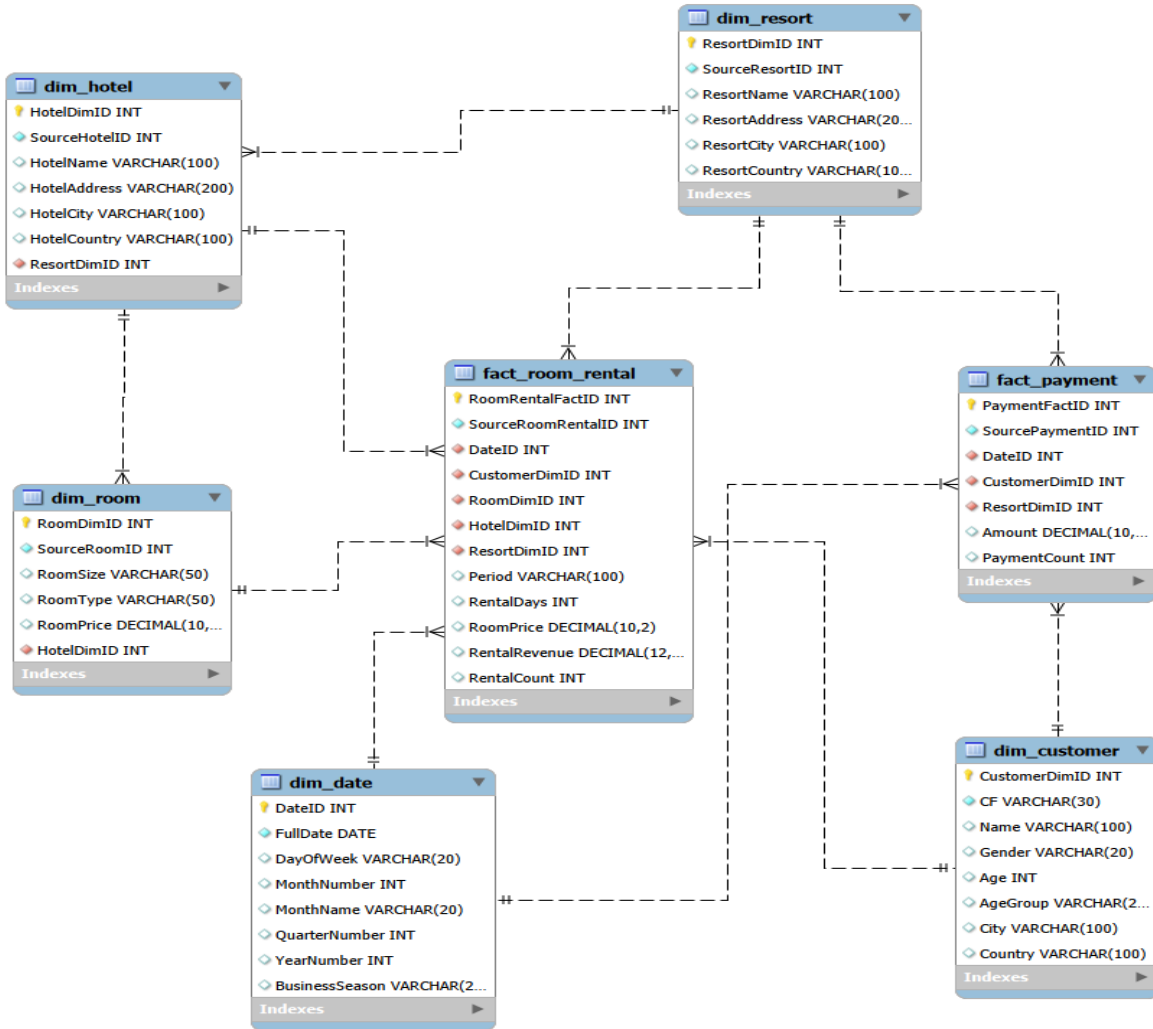
- Accommodation Performance Analysis
- Occupancy Analysis
- Hotel Performance Analysis
- Resort Performance Analysis
- Seasonal Rental Analysis



**Figure 7.4: ROOM\_RENTAL Data Mart**

## 7.7 Implemented Data Warehouse Schema

Figure 7.5 presents the implemented Data Warehouse schema after completion of the ETL process. The schema integrates the dimension and fact tables required for multidimensional analysis and Business Intelligence reporting.



**Figure 7.5: Data Warehouse Schema**

## 7.8 Summary

This chapter presented the implementation of the Data Warehouse and the SQL-based ETL process used to populate it. The resulting warehouse consists of five dimension tables and two fact tables supporting the Payment and ROOM\_RENTAL Data Marts. The implemented warehouse serves as the analytical foundation for the Business Intelligence environment presented in the next chapter.

## Chapter 8: Business Intelligence and Reporting

### 8.1 Overview

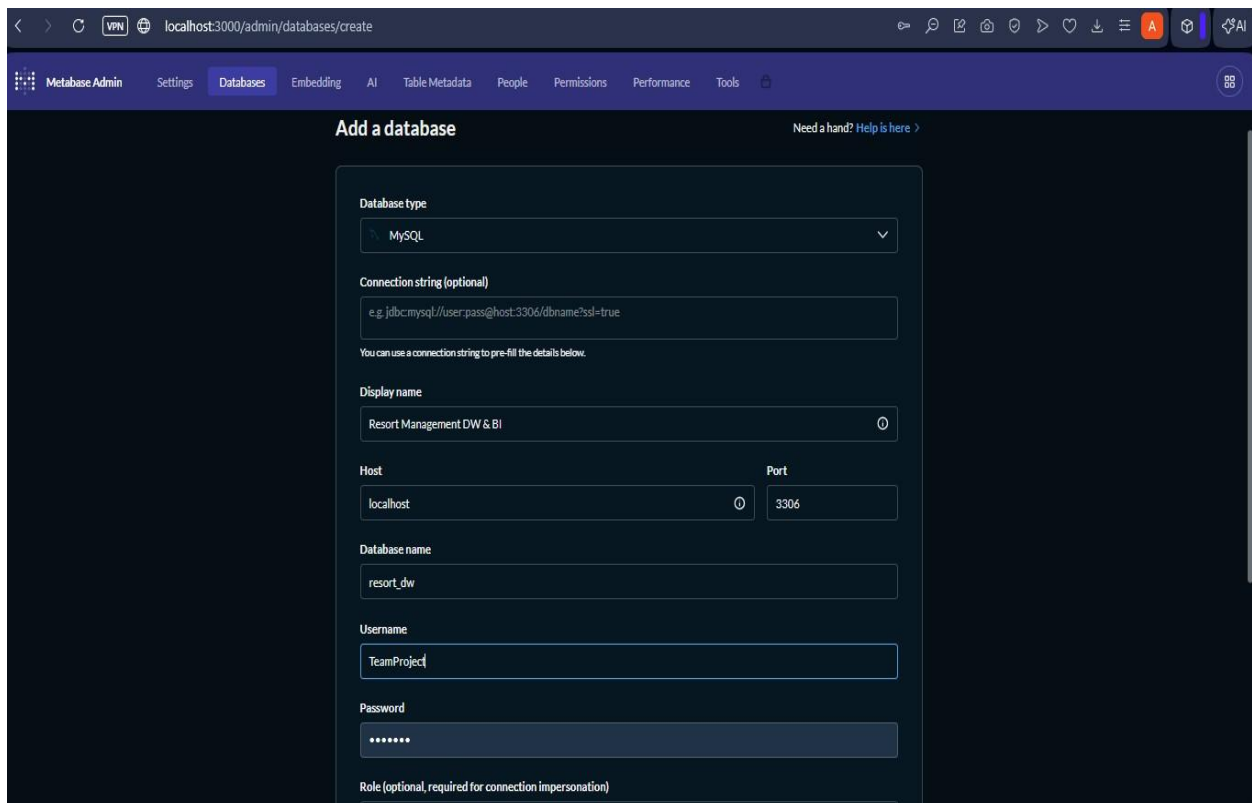
This chapter presents the Business Intelligence environment developed using Metabase. The platform was connected to the implemented Data Warehouse (resort\_dw) and used to create dashboards and analytical reports that support the decisional queries identified during the requirements analysis phase.

The Business Intelligence layer provides KPI monitoring, geographical analysis, demographic analysis, seasonal analysis, and ROOM\_RENTAL performance analysis through interactive visualizations and reports (Metabase, n.d.).

### 8.2 Metabase Configuration

Metabase was deployed locally using the Java JAR installation and connected directly to the resort\_dw Data Warehouse.

Figure 8.1 shows the configuration used to connect Metabase to the Data Warehouse.



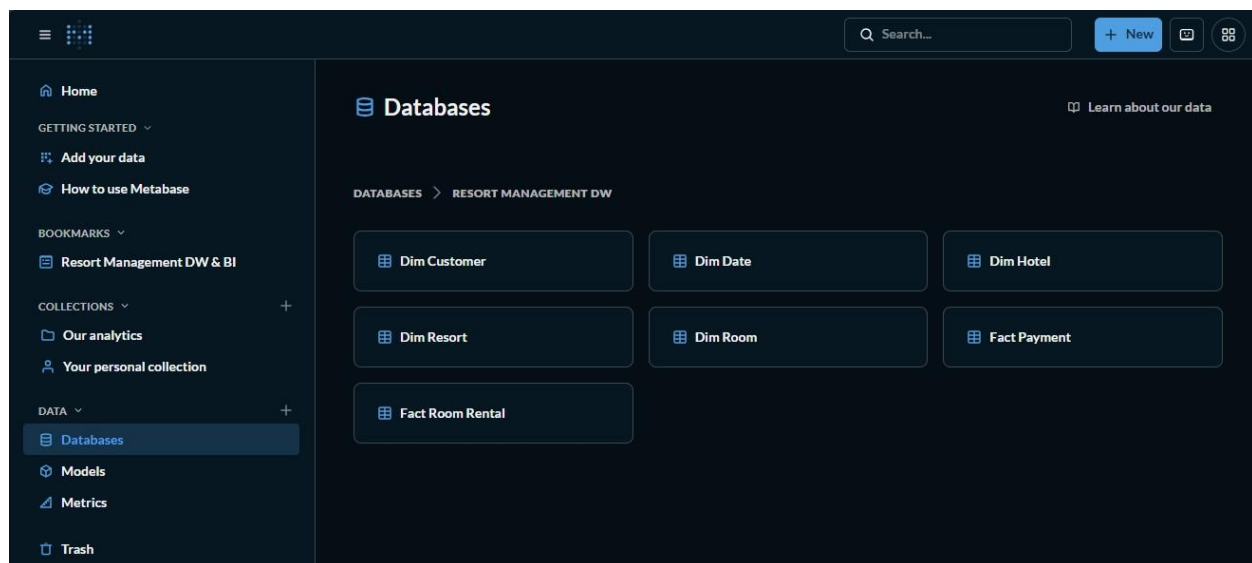
The screenshot displays the Metabase Admin interface at the URL `localhost:3000/admin/databases/create`. The navigation bar includes links for Metabase Admin, Settings, Databases, Embedding, AI, Table Metadata, People, Permissions, Performance, and Tools. The main content area is titled 'Add a database' and contains a form for configuring a new database connection. The form fields are as follows:

- Database type:** A dropdown menu set to 'MySQL'.
- Connection string (optional):** A text input field containing the example string `e.g. jdbc:mysql://user.pass@host:3306/dbname?ssl=true`. Below it, a note states: 'You can use a connection string to pre-fill the details below.'
- Display name:** A text input field containing 'Resort Management DW & BI'.
- Host:** A text input field containing 'localhost'.
- Port:** A text input field containing '3306'.
- Database name:** A text input field containing 'resort\_dw'.
- Username:** A text input field containing 'TeamProject'.
- Password:** A password input field with masked characters '\*\*\*\*\*'.
- Role (optional, required for connection impersonation):** An empty text input field.

**Figure 8.1: Metabase Data Source Configuration**



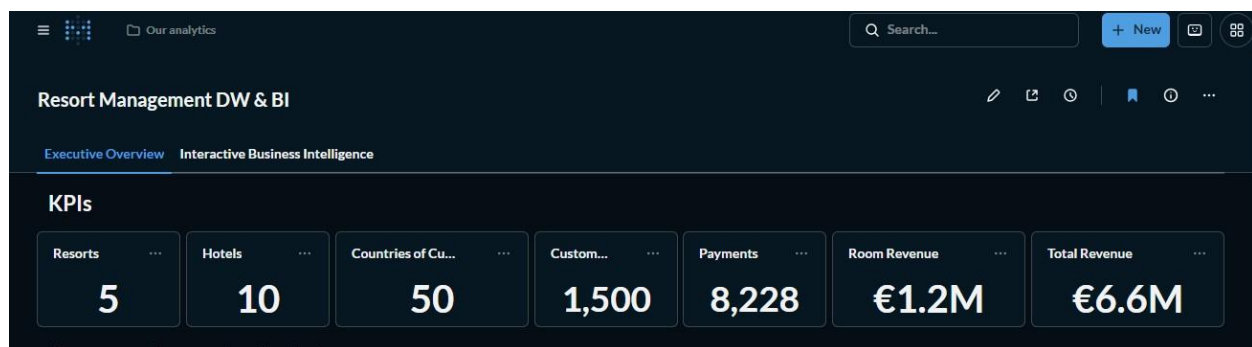
Figure 8.2 shows the implemented Data Warehouse tables available within Metabase after successful connection.



*Figure 8.2: resort\_dw Database in Metabase*

The connection enabled direct access to the dimension and fact tables implemented in the Data Warehouse and provided the foundation for dashboard and report development.

### 8.3 Executive Dashboard



*Figure 8.3: Executive Dashboard*

The Executive Dashboard provides a high-level overview of resort operations through KPI cards and summary visualizations.

The dashboard includes the following KPI cards:

- Total Resorts
- Total Hotels

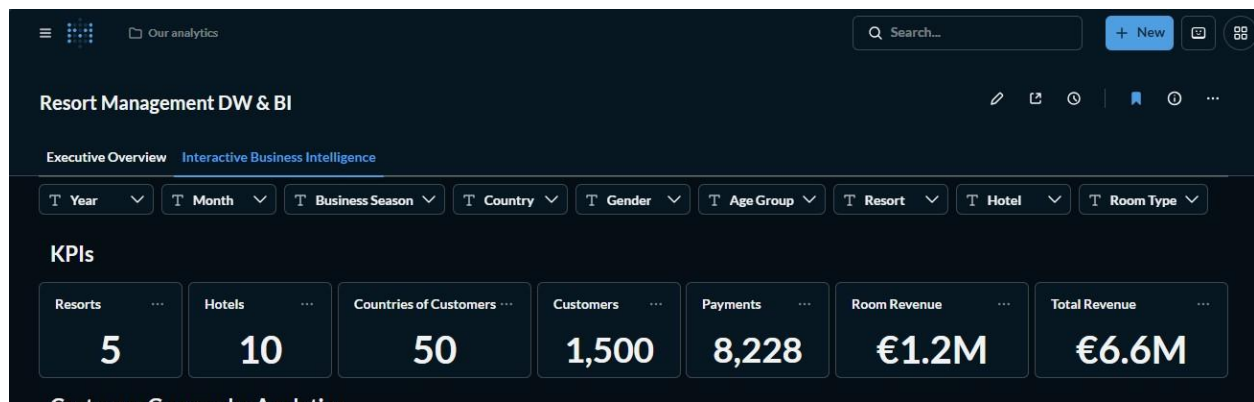
- Customer Countries
- Total Customers
- Total Payments
- Room Revenue
- Total Revenue

In addition to KPI monitoring, the dashboard contains visualizations grouped into the following analytical categories:

- Customer Geography Analytics
- Customer Demographics Analytics
- Season and Time Analytics
- ROOM\_RENTAL Analytics

These visualizations provide a consolidated view of business performance and support executive-level decision-making.

## 8.4 Interactive Dashboard



*Figure 8.4: Interactive Dashboard*

The Interactive Dashboard was developed to support multidimensional exploration of Data Warehouse information.

Unlike the Executive Dashboard, the Interactive Dashboard includes filters that allow users to dynamically modify the displayed results.

The available filters include:

- Year
- Month
- Business Season
- Country
- Gender

- Age Group
- Resort
- Hotel
- Room Type

These filters are connected to the dashboard visualizations and KPI cards, allowing users to analyse information from different perspectives and levels of detail.

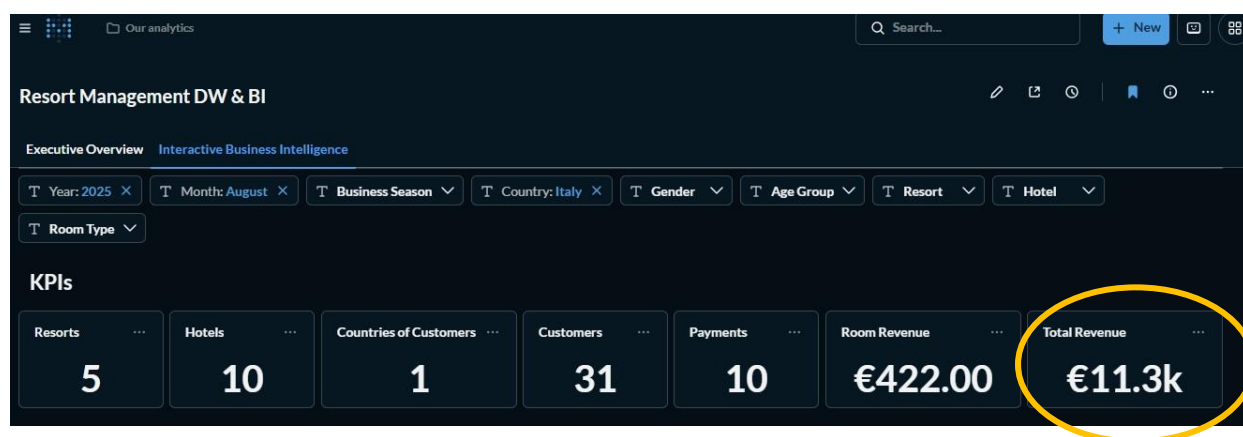
The Interactive Dashboard supports filter-driven multidimensional analysis across time, geography, customer demographics, accommodation characteristics, and organizational structures.

## 8.5 Analytical Reports

In addition to the dashboards, several analytical reports were developed to answer selected decisional queries and demonstrate the capabilities of the implemented Data Warehouse.

### 8.5.1 Revenue Analysis Report

As shown in Figure 8.5, the report directly supports **Query 1**, which examines revenue generated by customers from Italy during August 2025.



*Figure 8.5: Revenue Analysis Report*

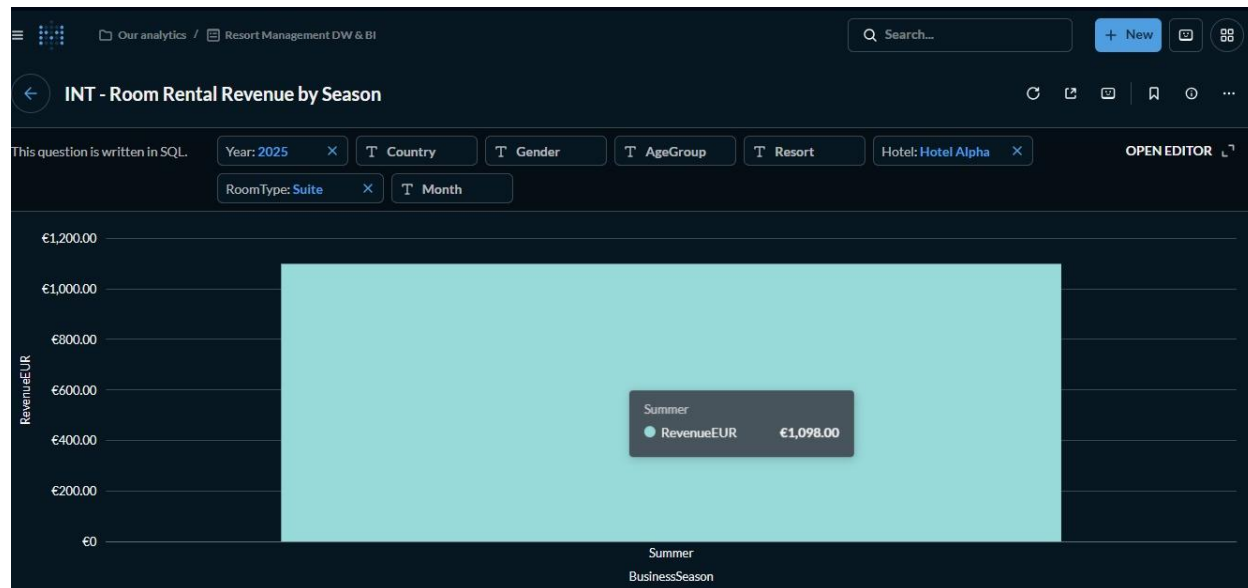
The Revenue Analysis Report was developed using the Payment Data Mart.

The report analyses revenue across customer geography and time dimensions using measures derived from FACT\_PAYMENT.

Managers can use this report to evaluate customer markets, identify revenue trends, and compare performance across different countries and seasons.

### 8.5.2 ROOM\_RENTAL Analysis Report

As shown in Figure 8.6, the report directly supports **Query 3**, which focuses on total rental income generated by Suite room types within Hotel Alpha during the Summer Business Season of 2025.



*Figure 8.6: ROOM\_RENTAL Analysis Report*

The ROOM\_RENTAL Analysis Report was developed using the ROOM\_RENTAL Data Mart. The report analyses rental revenue and accommodation performance across room, hotel, resort, and time dimensions using FACT\_ROOM\_RENTAL.

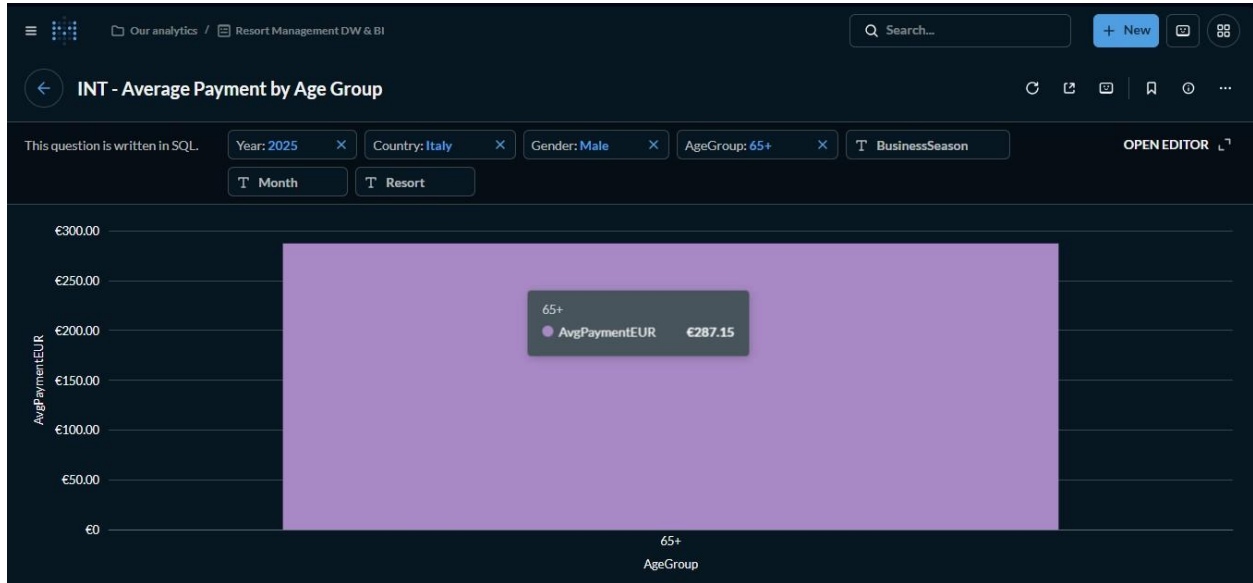
Managers can identify room types that contribute significantly to revenue and evaluate accommodation performance across different locations.

### 8.5.3 Customer Demographic Analysis Report

The Customer Demographic Analysis Report was developed using the Payment Data Mart. The report analyses payment behaviour across customer demographic and geographical dimensions using FACT\_PAYMENT.

As shown in Figure 8.7, the report directly supports **Query 8**, which examines the average payment amount for male customers aged 65 years and above from Italy during 2025.

Managers can examine payment patterns across different customer segments based on gender, age group, and country.



*Figure 8.7: Customer Demographic Analysis Report*

## 8.6 Decision Support Capabilities

The implemented Business Intelligence environment supports multidimensional analysis across revenue, accommodation performance, customer demographics, customer geography, and seasonal trends.

The Executive Dashboard provides a strategic overview of business performance, while the Interactive Dashboard supports detailed analytical exploration through dynamic filtering. Shared dimensions across the Payment and ROOM\_RENTAL Data Marts ensure consistency in reporting and support integrated analysis across multiple business processes.

## 8.7 Summary

This chapter presented the Business Intelligence environment implemented using Metabase. KPI cards, analytical visualizations, interactive dashboards, and analytical reports were developed using the Data Warehouse as the analytical data source. The resulting environment supports multidimensional analysis and managerial decision-making across the resort management domain.

## **Chapter 9: Challenges and Lessons Learned**

### **9.1 Overview**

This chapter presents the main challenges encountered during the project and the lessons learned throughout the design and implementation of the Resort Management Data Warehouse and Business Intelligence solution.

### **9.2 Challenges Encountered**

#### **9.2.1 Schema Reification**

One of the initial challenges was the transformation of the original E/R schema into a structure suitable for dimensional modelling. Several many-to-many relationships contained analytical information and required reification before conceptual modelling could begin.

#### **9.2.2 Fact Selection**

Multiple candidate facts were identified from the reified schema. Selecting the most suitable facts required analysing the decisional queries and determining which business processes provided the greatest analytical value. This resulted in the selection of PAYMENT and ROOM\_RENTAL for implementation.

#### **9.2.3 Dimensional Modelling**

Designing appropriate dimensions, measures, and hierarchies required careful alignment with the decisional queries. Pruning and grafting were necessary to remove non-analytical attributes and introduce analytical hierarchies suitable for multidimensional analysis.

#### **9.2.4 Data Generation and ETL**

The operational database required a sufficiently large and realistic dataset for analytical processing. Generating consistent synthetic data while maintaining referential integrity across entities and relationships required additional validation during both data generation and ETL implementation.

#### **9.2.5 Dashboard Design**

Developing dashboards that effectively supported the selected decisional queries required balancing summary-level reporting with detailed analytical exploration. Appropriate KPIs, visualizations, and filters had to be selected to support different analytical perspectives.

### 9.3 Lessons Learned

The project provided practical experience in the complete Data Warehouse development lifecycle, from requirements analysis to Business Intelligence reporting.

Key lessons learned include:

- The importance of selecting facts based on business requirements.
- The role of reification in preparing operational schemas for analytical modelling.
- The significance of dimensions, hierarchies, and measures in supporting multidimensional analysis.
- The importance of ETL processes in transforming operational data into analytical structures.
- The value of dashboards and reports in supporting data-driven decision-making.

The project also demonstrated the relationship between Operational Databases, Data Warehouses, Data Marts, and Business Intelligence platforms within an integrated analytical environment.

### 9.4 Summary

This chapter presented the main challenges encountered during the project and the lessons learned throughout the implementation process. These experiences contributed to a deeper understanding of Data Warehouse design, ETL development, dimensional modelling, and Business Intelligence reporting.

## Chapter 10: Conclusion and Future Work

### 10.1 Overview

This chapter concludes the Resort Management Data Warehouse and Business Intelligence project by summarizing the work completed, evaluating the objectives achieved, and identifying possible areas for future enhancement.

### 10.2 Project Summary

The project implemented a complete Data Warehouse and Business Intelligence solution based on the Resort Management E/R schema provided for the assignment.

The project began with requirements analysis and the identification of decisional queries. Candidate facts were identified, the E/R schema was refined through reification, and conceptual modelling was performed using attribute trees, pruning and grafting, dimensions, measures, and fact schemas.

The resulting designs were transformed into snowflake schemas and implemented within a MySQL-based Data Warehouse. A SQL-based ETL process was developed to transform operational data into analytical structures. Finally, Metabase was integrated with the Data Warehouse to provide dashboards and analytical reports that support managerial decision-making.

### 10.3 Objectives Achieved

The objectives defined for the project were successfully achieved.

Specifically:

- Decisional queries were identified from the Resort Management domain.
- Candidate facts were identified and analysed.
- The original E/R schema was refined through reification.
- Conceptual and logical designs were developed.
- The operational database was implemented and populated.
- The Data Warehouse was implemented using dimension and fact tables.
- ETL processes were developed and executed successfully.
- Data marts were implemented for PAYMENT and ROOM\_RENTAL.
- Business Intelligence dashboards and analytical reports were developed using Metabase.

These outcomes demonstrate the successful implementation of the complete Data Warehouse development lifecycle.

### 10.4 Future Work

Several enhancements may be considered in future implementations:

- Implementation of additional data marts based on the remaining candidate facts.
- Integration with real operational resort data instead of synthetic datasets.



- Automation of ETL processes through scheduled execution.
- Deployment of the solution within a cloud-based environment.
- Development of predictive analytics and forecasting models.
- Expansion of the Business Intelligence environment with additional dashboards and analytical reports.

### **10.5 Closing Remarks**

The implemented solution demonstrates how operational data can be transformed into meaningful analytical information through Data Warehouse technologies and Business Intelligence tools. The resulting environment provides a structured foundation for multidimensional analysis and supports evidence-based decision-making within the resort management domain.

### **Team Members Contribution**

All team members actively participated throughout the project lifecycle and contributed to the successful completion of the work.

Activities including requirements analysis, decisional query development, schema reification, conceptual modelling, logical design, operational database implementation, Data Warehouse implementation, ETL development, dashboard creation, report preparation, testing, and presentation were carried out collectively through continuous planning, discussion, and coordination.

The team therefore considers the contributions of all members to be substantially equal.

## References

- Golfarelli, M., & Rizzi, S. (2009). *Data warehouse design: Modern principles and methodologies*. McGraw-Hill.
- Inmon, W. H. (2005). *Building the data warehouse* (4th ed.). Wiley.
- Metabase. (n.d.). *Metabase documentation*. <https://www.metabase.com/docs>

## Appendix A: GitHub Repository Information

The complete implementation of the Resort Management Data Warehouse and Business Intelligence project is available in the project GitHub repository.

The repository contains all source code, database scripts, Data Warehouse scripts, dashboard exports, diagrams, and project documentation used during the implementation.

Repository URL:

<https://github.com/FrancisOfosu/resort-management-dw-bi>

Repository Structure:

Resort\_Management\_DW\_BI/

├── Python\_Data\_Generation/

├── OLTP\_Database/

├── Data\_Warehouse/

├── Metabase\_Reports/

├── Diagrams/

├── Documentation/

└── README.md

## Appendix B: Sample Generated Data

```
1 • SELECT 'DIM_CUSTOMER' AS TableName, COUNT(*) AS TotalRecords FROM DIM_CUSTOMER
2 UNION ALL
3 SELECT 'DIM_DATE', COUNT(*) FROM DIM_DATE
4 UNION ALL
5 SELECT 'DIM_RESORT', COUNT(*) FROM DIM_RESORT
6 UNION ALL
7 SELECT 'DIM_HOTEL', COUNT(*) FROM DIM_HOTEL
8 UNION ALL
9 SELECT 'DIM_ROOM', COUNT(*) FROM DIM_ROOM
10 UNION ALL
11 SELECT 'FACT_PAYMENT', COUNT(*) FROM FACT_PAYMENT
```

Result Grid   Filter Rows:  Export:  Wrap Cell Content: 

|   | TableName        | TotalRecords |
|---|------------------|--------------|
| ▶ | DIM_CUSTOMER     | 1500         |
|   | DIM_DATE         | 1086         |
|   | DIM_RESORT       | 5            |
|   | DIM_HOTEL        | 10           |
|   | DIM_ROOM         | 100          |
|   | FACT_PAYMENT     | 8228         |
|   | FACT_ROOM_RENTAL | 900          |

## Appendix C: ETL Mapping Summary

This appendix summarizes the mapping between the Operational Database and the Data Warehouse during the ETL process.

**Table C.1: Dimension Mapping**

| Operational Source | Data Warehouse Target |
|--------------------|-----------------------|
| PERSON             | DIM_CUSTOMER          |
| RESORT             | DIM_RESORT            |
| HOTEL              | DIM_HOTEL             |
| ROOM               | DIM_ROOM              |
| PAYMENT.Date       | DIM_DATE              |
| ROOM_RENTAL.Date   | DIM_DATE              |

**Table C.2: Fact Mapping**

| Operational Source | Data Warehouse Target |
|--------------------|-----------------------|
| PAYMENT            | FACT_PAYMENT          |
| ROOM_RENTAL        | FACT_ROOM_RENTAL      |

**Table C.3: Derived Analytical Attributes**

| Data Warehouse Attribute | Source             | Transformation                   |
|--------------------------|--------------------|----------------------------------|
| Business Season          | Date               | Derived from month               |
| Day of Week              | Date               | Derived from transaction date    |
| Month Number             | Date               | Derived from transaction date    |
| Month Name               | Date               | Derived from transaction date    |
| Quarter Number           | Date               | Derived from transaction date    |
| Year Number              | Date               | Derived from transaction date    |
| Age Group                | PERSON.Age         | Categorization into age ranges   |
| RentalDays               | ROOM_RENTAL.Period | Conversion to numerical duration |
| RentalRevenue            | ROOM_RENTAL, ROOM  | Room Price $\times$ RentalDays   |

## Appendix D: Additional Dashboard Screenshots

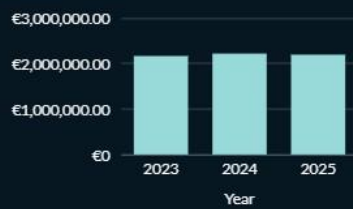
This appendix contains supplementary dashboard and report screenshots that were not included in Chapter 8.

The screenshots provide additional examples of the analytical capabilities implemented within the Metabase Business Intelligence environment.



## Season/Time Analytics

Revenue by Year



Monthly Revenue Trend



Revenue by Business Season

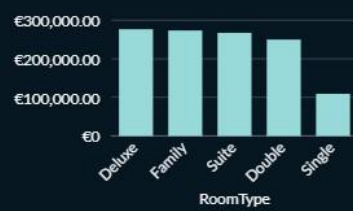


Average Payment by Season



## Room Rental Analytics

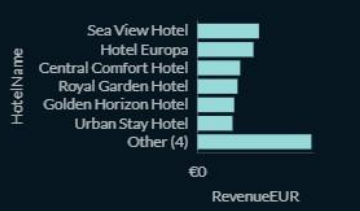
Revenue by Room Type



Rental Count by Room Type



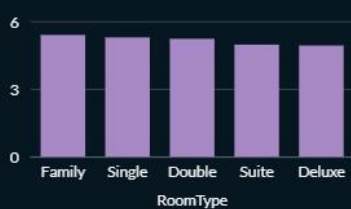
Revenue by Hotel



Revenue by Resort



Average Length of Stay by Room Type



Room Rental Revenue by Season



Average Revenue per Rental by Room T...

